



Topic: Relational Algebra

Exercise 1

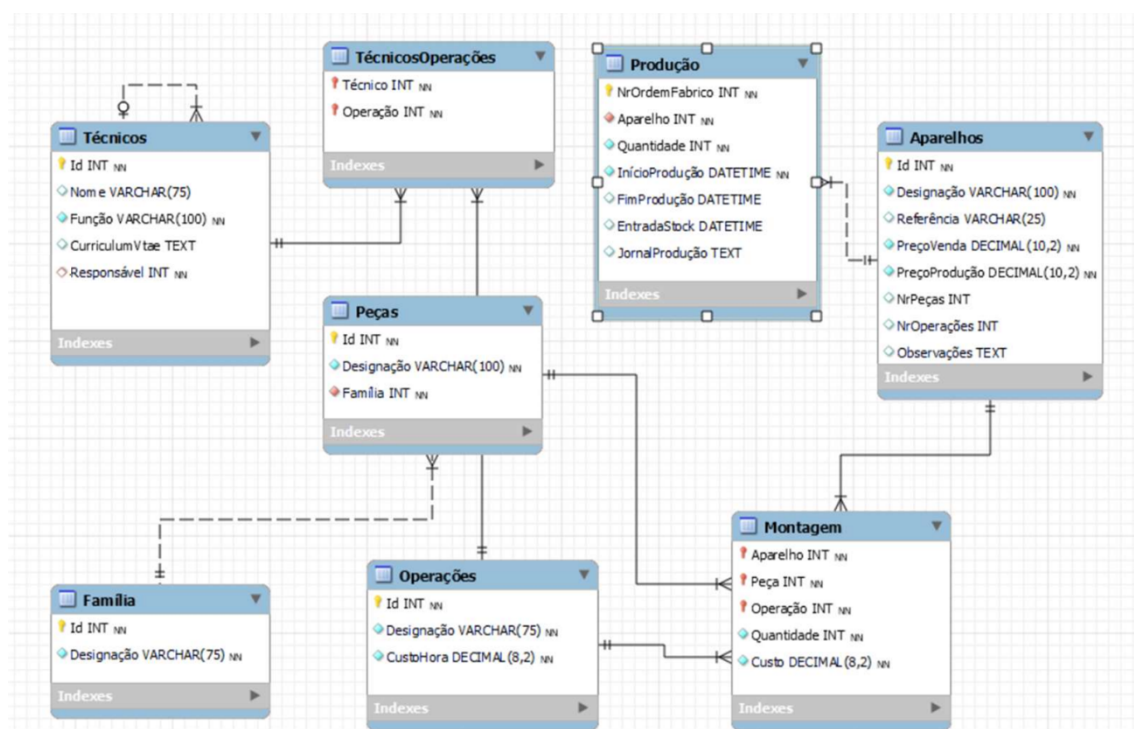


Figure 1 – Logical schema. (Belo, O.)

Consider the logical scheme presented in Figure 1, relating to a database of a company that assembles appliances. After carefully and in detail the diagram presented, answer the following exercises using relational algebra with the respective trees.

1. Which are the devices with a selling price greater than €100?
2. What are the names of the technicians with the role "Reviewer"?
3. Which are the parts belonging to family 1?
4. Which are the operations with an hourly cost greater than or equal to €20?
5. Which are the devices with selling price > €100 and number of parts less than 10?
6. Which are the production records started in the years 2025 and 2024?
7. What are the names of the technicians who performed operations?
8. What are the names (designations) of the devices that use part 1?
9. What are the names (designations) of the parts from family 1 used in assemblies with cost greater than €50?

10. What are the names of the technicians who worked on operations with cost > €20?
11. Which are the devices produced after 01/01/2025 with quantity > 50?
12. Which are the technicians who worked on devices with price > €200?
13. Which are the devices that do NOT have any recorded production?
14. Which are the technicians who did NOT perform any operation?
15. Which are the operations common to both assembly and technicians?
16. What are the names of the technicians who worked on operations with cost > €20 on devices with price > €100?
17. Which are the devices that use parts from family 1 and also from family 2?
18. Which are the technicians who did NOT work on operations with cost > €30?
19. How many parts exist?
20. Which are the devices whose total produced quantity is greater than 100?
21. What is the average cost of assemblies for each part, showing the parts with higher averages first?
22. Which are the technicians who participated in assemblies of devices whose total production is greater than 100?
23. Which are the devices whose total assembly cost is greater than the average cost?